

Construction + Validation Doubles (next to build)

2026.07.20 - Doubles for BOTH Triple Reconstruction and Assay Validation

Script: experiments/010-kuzmin-tmi/scripts/construction_validation_doubles.py Data: experiments/010-kuzmin-tm

Why the pure set-cover was the wrong objective

The essential wet-lab task is a small set of double mutants that (a) reconstructs many top-ranked triples AND (b) has **variance in DMF / interaction** to validate the new echo-plating assay against (vs plate-stamping). The triple-coverage set-cover ([[experiments.010-kuzmin-tmi.scripts.optimized_doubles_setcover_constructed_10]]) does (a) with 8 doubles but fails (b): those 8 are all **near-neutral** — Costanzo DMF span 0.15, $|\varepsilon| \leq 0.039$, and **zero significant interactions** — so the assay would have almost nothing to discriminate. An assay validated only on near-1.0 fitness and null interactions has not been shown to detect signal.

Two tiers, unioned (13 doubles)

- **coverage (8)** — the greedy set-cover doubles; reconstruct **all 31** within-10 top-k triples.
- **validation (5)** — added for dynamic range + real signal: the doubles with a **significant Costanzo interaction** ($P < 0.05$ & $|\varepsilon| > 0.08$, both signs) plus the **lowest- and highest-DMF** doubles.

Result: 31/31 triples still reconstructed, DMF range **0.513–1.178** (span 0.665, $4.4\times$ the coverage-only 0.152), ε from **−0.130 to +0.098**, **3 significant interactions**.

Gene reuse across the 13 doubles (most-reused first)

Each of the 10 constructed genes appears in this many of the 13 doubles; the two hubs (YOS9/YDR057W and ELC1/YPL046C) carry most of the triple coverage.

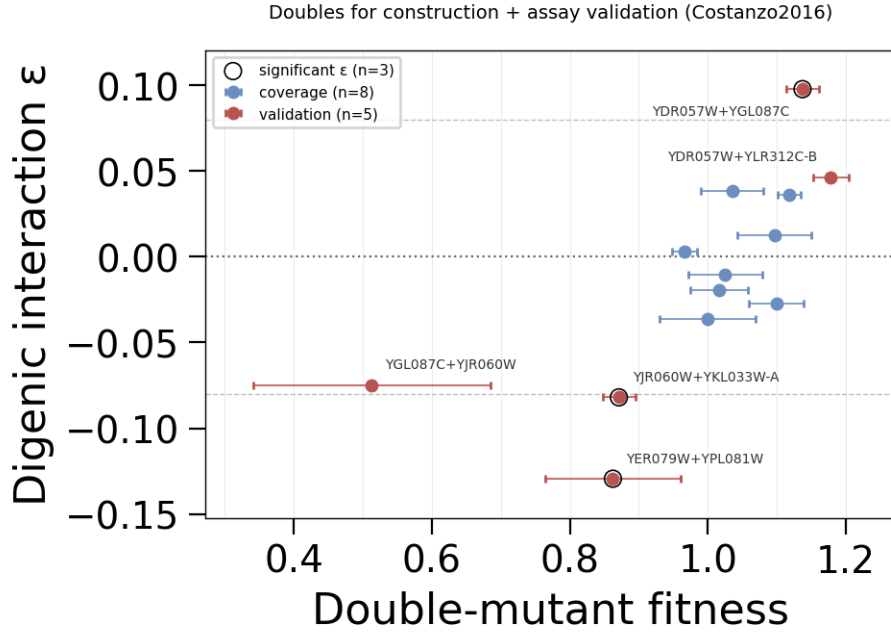
gene (systematic)	common	in # doubles
YDR057W	YOS9	5
YER079W	—	3
YJR060W	CBF1	3
YLR312C-B	—	3
YPL046C	ELC1	3
YPL081W	RPS9A	3
YGL087C	MMS2	2
YLL012W	YEH1	2
YBR203W	COS111	1
YKL033W-A	—	1

The 13 doubles, ordered by gene reuse

Ordered so the doubles built from the most-reused genes come first (reuse Σ = the sum of the two genes' double-counts). Names are systematic ORF with common name in parentheses where one exists.

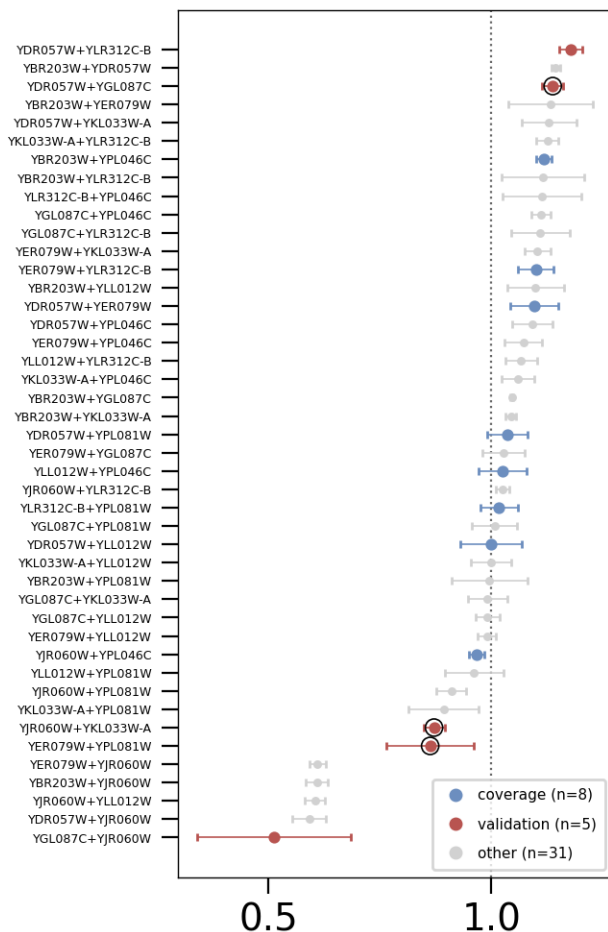
#	double (systematic / common)	reuse Σ (g1+g2)	tier	DMF $\pm$ SD	ε	p	sig	# triples
1	YDR057W (YOS9) + YPL081W (RPS9A)	8 (5+3)	cov	1.035 ± 0.045	+0.039	0.260		3
2	YDR057W (YOS9) + YER079W	8 (5+3)	cov	1.096 ± 0.054	+0.012	0.435		5
3	YDR057W (YOS9) + YLR312C-B	8 (5+3)	val	1.178 ± 0.026	+0.047	0.119		1
4	YDR057W (YOS9) + YLL012W (YEH1)	7 (5+2)	cov	1.000 ± 0.069	−0.036	0.296		5
5	YDR057W (YOS9) + YGL087C (MMS2)	7 (5+2)	val	1.137 ± 0.024	+0.098	0.001	✓	3
6	YER079W + YPL081W (RPS9A)	6 (3+3)	val	0.863 ± 0.098	−0.130	0.035	✓	0
7	YJR060W (CBF1) + YPL046C (ELC1)	6 (3+3)	cov	0.966 ± 0.018	+0.003	0.453		4
8	YLR312C-B + YPL081W (RPS9A)	6 (3+3)	cov	1.016 ± 0.042	−0.019	0.355		3
9	YER079W + YLR312C-B	6 (3+3)	cov	1.099 ± 0.039	−0.027	0.290		5
10	YGL087C (MMS2) + YJR060W (CBF1)	5 (2+3)	val	0.513 ± 0.172	−0.075	0.268		0
11	YLL012W (YEH1) + YPL046C (ELC1)	5 (2+3)	cov	1.025 ± 0.053	−0.010	0.429		5

#	double (systematic / common)	reuse Σ (g1+g2)	tier	DMF $\pm$ SD	ε	p	sig	# triples
12	YJR060W (CBF1) + YKL033W-A	4 (3+1)	val	0.871 ± 0.023	-0.082	0.007	✓	0
13	YBR203W (COS111) + YPL046C (ELC1)	4 (1+3)	cov	1.118 ± 0.017	+0.036	0.077		5



All 44 measured within-10 doubles ranked by DMF, with the 13 selected marked by tier (coverage = blue (C), validation = red (V), unused = gray; significant ε ringed) — the tiered companion to `constructed_10_dmf_forest` (which predates the validation tier and marks only the 8 set-cover doubles):

All 44 measured within-10 doubles by DMF
color = tier; ring = significant ϵ



Double-mutant fitness (Costanzo2016)

Notes: DMF/SD/SE and ϵ/p are the published Costanzo values (SD = sample SD over 4 colonies, directly comparable to the assay's colony SD — see [[experiments.010-kuzmin-tmi.scripts.constructed_10_dmf_reference]]). The low-DMF end is dominated by CBF1/YJR060W (its single defect), so a low DMF there is a fitness effect, not an interaction. Two of the three significant doubles (YER079W+YPL081W, YJR060W+YKL033W-A) enable no top-k triple — they are pure validation targets; the third (YDR057W+YGL087C, strongest positive ϵ) also re-builds 3 triples. The set is tunable: drop the two triple-less validation doubles for an 11-double set, or add more near-significant ϵ to stress the assay.

There are only 3 significant interactions in the whole panel

Surveyed across all 66 panel-12 doubles and all three sources: exactly **3** clear Costanzo's bar ($P < 0.05$ & $|\epsilon| > 0.08$), and all 3 are within-10 and already in the validation tier — YDR057W+YGL087C (+0.098), YJR060W+YKL033W-A (−0.082), YER079W+YPL081W (−0.130). None hide in the dropped-gene doubles; Kuzmin2018/2020 have too few measured pairs to reach significance. So “construct the significant ones” = these 3, already selected. The panel is near-neutral by design, so strong interactions are scarce.

High- $|\epsilon|$ but insignificant doubles (e.g. YGL087C+YJR060W, $\epsilon = -0.075$, $p = 0.27$, DMF SD 0.172) arise from **large SE(ϵ)**: ϵ significance is $|\epsilon|/\text{SE}(\epsilon)$, and $\text{SE}(\epsilon)$ folds in the SD of the double-mutant fitness AND both single-mutant fitnesses ($n=4$ colonies each). A wide colony SD or a noisy constituent single (CBF1, RPS9A) inflates $\text{SE}(\epsilon)$ so even a sizable ϵ is indistinguishable from the multiplicative null. Mirror case: YBR203W+YDR057W ($\epsilon = +0.062$, below the magnitude bar) is highly significant ($p = 1.4e-5$) because its DMF SD is 0.010. For assay validation the 3 *significant* doubles are the strong anchors; high-SD/high- $|\epsilon|$ doubles validate fitness dynamic range, not interaction

detection.

Sampling depth beats SD — the assay’s key lever. The resolving quantity is SE, not SD: $SE = SD/\sqrt{n}$. SD is a fixed biological property of a strain (colony-to-colony spread), but **n is ours to choose** — so a “high-SD, insignificant” double is not disqualified, it just needs more colonies. Because our goal is **sampling specific predictions precisely (depth), not screening many strains (breadth)**, the right trade is few doubles $\times$ many colonies each: plate a high-SD double like YGL087C+YJR060W (SD 0.172) at large n and its SE shrinks by $\sqrt{n}$ until the interaction resolves. This is the opposite of a genome-wide screen (breadth, low n per pair) and is what echo dispensing enables — dense, uniform replication of a handful of constructs. So the validation tier is chosen for interesting *point estimates* (DMF extremes, high $|\epsilon|$); driving each to significance is a sampling decision at the bench, not a property of the reference. Caveat: deep sampling lowers *our* SE, but the Costanzo reference stays n=4 — so once our error bar is tighter than theirs, a disagreement may reflect the reference’s own noise as much as the assay’s, making our measurement the more trustworthy of the two.

Related: `[[experiments.010-kuzmin-tmi.scripts.topk_triples_from_constructed_10]]`, `[[experiments.010-kuzmin-tmi.scripts.optimized_doubles_setcover_constructed_10]]`.